

Positive impacts of rangeland abandonment on dung beetle functional diversity

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ABSTRACT

We assessed the effect of rangeland abandonment on dung beetle biodiversity and functional diversity in an African semi-desert region. Dung beetles were sampled on farms that were abandoned either over 10 years ago, abandoned one year ago, or not abandoned. We sampled from tree dominant landscape features including dry riverbeds, flatlands, and hills. We hypothesised that rangeland abandonment would decrease dung beetle biodiversity and functional diversity, also testing whether this pattern holds true across the dominant landscape features. Preference experiments revealed that all dung beetle species preferred dung resources other than sheep dung. Contrary to expectations, dung beetle biodiversity and functional richness was higher on abandoned rangelands. The landscape features that were particularly significant in sustaining high dung beetle diversity and functional diversity in this ecosystem were the flatlands and dry riverbeds, the main corridors of movement of animals. We speculate that rangeland abandonment improved conditions for most dung beetle species, due to decreased pressures on remaining native mammal populations. Despite this, larger species that need greater dung resources for development became rare after abandonment. Reintroductions of native mammals should therefore be encouraged to promote dung beetle biodiversity and functional diversity in this semi-desert ecosystem.

1. Introduction

The removal of livestock without replacement with native herbivores alters grazing regimes, with cascading, often negative effects on other biota (Charles, 2018). In addition, when abandoned land is incorrectly managed, the negative impacts on biodiversity can be severe due to a wide variety of influences, including increases in populations of invasive species, altered fire regimes, reduced water availability, and increased bush encroachment (Munroe et al., 2013). Alternatively, rangeland abandonment may reduce pressures on remaining native mammal populations and their predators, with positive effects on other biota (Li and Li, 2017). The outcomes for native biodiversity on abandoned rangelands are therefore unpredictable and largely dependent on proper post-abandonment management.

Rangeland abandonment leads to a loss of dung resources, which can negatively affect keystone taxa, such as dung beetles, that are directly dependent on this resource (Nichols et al., 2009). This close association of dung beetles with mammals, combined with their sensitivity to alterations in environmental conditions, makes them ideal bioindicators of ecological disturbance and in landscape ecology (McGeoch et al.,

2002; Nependa et al., 2021; Pryke et al., 2016; Spector, 2006). They are also directly responsible for many fundamental ecosystem processes, such as parasite and pest control, seed dispersal, nutrient cycling, and vegetation development, making them essential parts of many terrestrial ecosystems (Nichols et al., 2008).

Ecosystem services, processes, and resilience are closely linked to functional diversity (Díaz et al., 2007; Petchey and Gaston 2002) and changes in functional diversity are believed to affect ecological processes more than changes in species richness (Hillebrand and Matthiessen, 2009; Hooper et al., 2005). This is because human activities result in changes in abundances of species with particular functional traits, effectively reducing or removing some ecosystem functions (Barragán et al., 2011). When monitoring alterations to an environment, studies should therefore ideally include the diversity of ecological roles and functional traits of the different species within the ecosystem (Berragan et al., 2011; Díaz et al., 2007). Most studies that investigated the influence of land-use change in terms of functional diversity have only focused on intensification processes. Rangeland abandonment, which is opposite to intensification, has received far less attention (e.g., MacDonald et al., 2000; Pakeman and Marriott, 2010; Tonelli et al.,

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2018).

Rangeland abandonment may occur due to changes in global markets or mismanagement of resources (Gellrich and Zimmermann, 2007) but may also follow strategic developments. Pertaining to the latter, a recent large-scale example is the abandonment of extensive areas of semi-desert Nama-Karoo rangeland in South Africa for the development of the Square Kilometre Array (SKA) project, as part of the world's largest radio telescope (Taylor, 2012). Consequently, after centuries of farming, ca. 13,000 sheep are no longer grazing the ca. 130,000 ha of land used for the SKA development. The impacts of rangeland abandonment on native biota and ecosystem functions in this semi-desert are unknown, but especially relevant as the often-overexploited, arid regions of the world take decades to recover after disturbances (Allen et al., 1995; Steed et al., 2018).

Dung beetles display a large diversity of behavioural and morphological traits that are linked to their multiple ecological roles (Davis et al., 2020). The present study, therefore, sets out to not only examine the effect of prominent semi-desert landscape features and rangeland abandonment on dung beetle richness, abundance, biomass, and community assemblage composition, but also to quantify changes in their functional diversity using a number of species traits linked to ecosystem function. We test the hypotheses that dung beetle biodiversity and functional diversity will differ between different semi-desert landscape features (dry riverbeds, flatlands, and rocky hills) and that rangeland abandonment, through removal of livestock without active replacement with wild grazers, will have negative effects on dung beetle biodiversity and functional diversity, as was found in other systems studied to date (Liberal et al., 2011; Lobo et al., 2006; Tonelli et al., 2018; Verdú et al., 2007).

2. Materials and methods

2.1. Study area

The Northern Cape Province of South Africa is dominated by the Nama-Karoo biome, a semi-desert ecosystem that consists mainly of a mosaic of flat sandy or stony plains, shale and dolerite hills, and ephemeral rivers (Mucina et al., 2006). Annual rainfall ranges between 100 mm and 500 mm and falls mostly in summer (Mucina et al., 2006). Native meso-herbivores historically included large migrating herds of wildebeest (*Connochaetes taurinus*), blesbok (*Damaliscus dorcas*), zebra (*Equus quagga*), eland (*Taurotragus oryx*) and springbok (*Antidorcas marsupialis*) (Lovegrove and Siegfried, 1993), providing an array of dung types (pelleted and otherwise) and consistencies (moist vs. dry). These, and many other animals, have largely been replaced by farming with domestic sheep, providing a homogenous dung resource (small pellets). Low numbers of native meso-herbivore species, such as kudu (*Tragelaphus strepsiceros*), and smaller herbivores, such as porcupine (*Hystrix africaeustralis*), grey duiker (*Sylvicapra grimmia*), and steenbok (*Raphicerus campestris*), remain on farms (Woodgate et al., 2018). In addition, farms still have numerous smaller omnivores present, such as various mongoose species, honey badger (*Mellivora capensis*), and baboon (*Papio ursinus*), as well as carnivores, such as jackal (*Lupulella mesomelas*) and caracal (*Caracal caracal*). The latter two are extensively hunted to protect livestock (Somers et al., 2018).

The SKA (Square Kilometre Array) radio astronomy observatory is situated ca. 90 km northeast from the small town of Carnarvon. To fulfil radio-silence and other regulatory and safety requirements, large areas surrounding the radio telescopes have been bought and the livestock subsequently removed. Farmland abandonment in the core of the SKA began in 2007 and the remainder of the land necessary to complete the project was acquired in 2017. The area, therefore, comprises a mosaic of

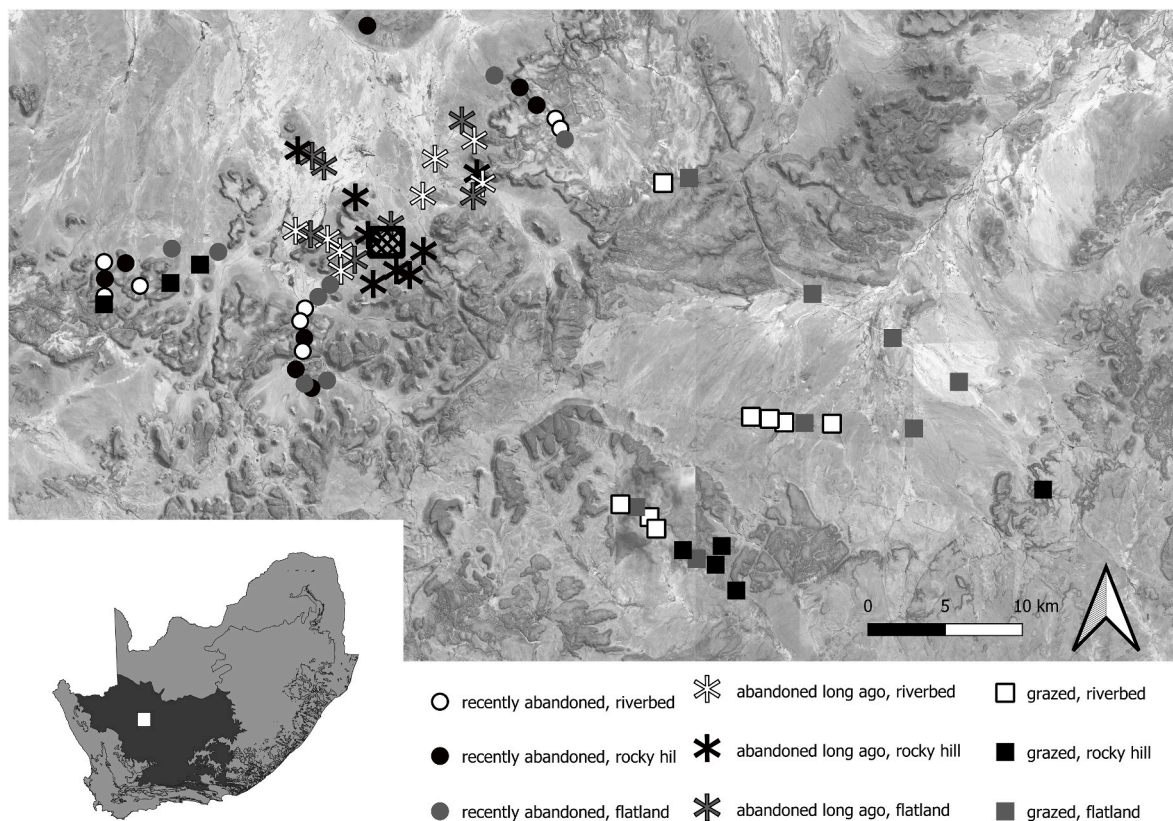


Fig. 1. Map of the study area containing the 72 sampling plots. The checkered block indicates the centre of the core area. Insert to figure represents a map of South Africa with the Nama-Karoo biome shaded in black and the sampling area indicated with a white block.

land-use practices, the core of which was abandoned by livestock farming more than 10 years ago, surrounded by recently abandoned farms (abandoned one year prior to this study) in a matrix of land still used for livestock farming (Fig. 1).

2.2. Sampling design and dung beetle collection

Sampling was conducted twice, once during spring (August–September 2018) and once during winter (June 2019). Three different levels of land abandonment were selected within the study area, including areas abandoned a long time ago (10 years or longer before first sampling), areas recently abandoned (one year prior to first sampling) and areas currently grazed by livestock. Within each of these different land use areas, 24 study plots were selected, each separated by at least 1 km to reduce the effect of pseudo replication (da Silva and Hernández, 2015b). These 24 plots consisted of eight sites, one in each of the three predominant landscape features, including plains/flatlands with sandy loam to coarse grains soils, rocky hills with loam soils (where all plots were placed on the cooler and mesic southern slopes), and dry riverbeds with coarse sand.

At each plot, four pitfall traps were placed in a square with at least 100 m separation to minimize trap interference between 08:00 and 12:00 in the morning (Larsen and Forsyth, 2005). Therefore, sampling effort for this study was: 3 x grazing regimes x 3 landscape features x 8 plots x 4 traps x 2 seasons = 476 traps in total. Traps consisted of plastic cups with openings of 5.7 cm and depths of 9.0 cm that were buried with their rims level with the soil surface (Pryke et al., 2013). As different taxa are attracted to different sources (Davis et al., 2020), each trap per plot was baited with a different bait source. Baits included pig dung (omnivore), cow dung (pats), sheep dung (pellets), and rotten chicken livers (carrion) (Pryke et al., 2013). Dung was collected from abattoirs and thoroughly mixed to create consistency in volatiles. Bait balls consisted of 150 g of dung or carrion that were placed into separate breathable material bags, frozen to kill pathogens and to ensure freshness and consistency in dung volatiles between different sampling days and suspended over the pitfall traps using wire (Pryke et al., 2016).

Traps were half filled with soapy water to capture beetles and left operational for 24 h. All individuals collected were frozen, separated into species, counted, and assigned to one of three functional guilds (paracoprid = tunnellers, endocoprid = dwellers, or telocoprid = rollers) based on their nesting behaviour (Davis et al., 2020). Ten randomly chosen individuals of each species (when available, otherwise all individuals) were dried and weighed to calculate the mean dry mass per species and the total mass of all individuals collected in each trap. A reference collection is housed in the Stellenbosch University Entomological Collection, Stellenbosch, South Africa.

2.3. Statistical analyses

2.3.1. Effects of dung type, landscape features and rangeland abandonment on dung beetle richness, abundance, biomass and community assemblage composition

Data for dung beetle catches for the two seasons were combined per bait type per plot prior to analyses. The carrion bait did not attract any dung beetle species in either sampling season. We therefore removed this bait type from further statistical analyses. Two non-parametric species estimators (Chao2 and Jackknife2) were calculated using PRIMER 6 (PRIMER-e, Auckland, New Zealand, 2016) to assess sampling coverage for overall assemblages and those separated by land use, dung type, and landscape feature (Colwell and Coddington, 1994). The influence of land use, dung type, and landscape feature (as fixed effects) on dung beetle species richness, abundance, and biomass were tested using generalized linear mixed-effect models (Bates et al., 2014) in R (R Development Core Team, 2020). Sites were categorized into four wind directions (NW, NE, SE and SW), according to their geographic location relative to the centre of the core region and followed a gradient of

increasing rainfall from north to south and east to west in the study region (Palmer et al., 1997). This was used as a spatial random variable to improve the model parameters. Dung beetle abundance and species richness data sets best fitted a Poisson distribution by a Laplace approximation and biomass data best fitted a gamma distribution with log link function (Bolker et al., 2009). Significant main effects were separated using conservative Tukey post-hoc tests in the *multcomp* package in R (Hothorn et al., 2008).

Effects of the environmental variables on dung beetle community assemblage composition were determined with 9999 permutations of permutational multivariate analyses of variance (PERMANOVA) on Bray–Curtis similarity matrices constructed from square-root-transformed abundance data (to reduce the effect of common species) in PRIMER 6. In addition, data were visualized using CAP analyses in PRIMER 6. Significant groupings based on PERMANOVA were further analysed using post-hoc pairwise tests in PRIMER 6.

2.3.2. Effects of landscape features and rangeland abandonment on dung beetle functional diversity

Ten traits were selected to define the functional identity of the species collected here. In addition to (1) nesting strategy (dung relocation behaviour) and (2) dry mass (as proxy for the dung mass needed for normal larval development) we also measured (3) the total length of the beetle (using digital callipers) from the tip of the abdomen to the tip of the clypeus, as larger beetles can manipulate more dung than smaller individuals (Nervo et al., 2014). (4) Activity was recorded from the literature (Davis et al., 2020) as nocturnal or diurnal (coded as 1—diurnal or 0—nocturnal for analyses). Diet preferences of the dung beetles in the region can be separated into coprophagous (dung), necrophagous (carrion), or generalists that will utilise more than one source (Davis et al., 2020). Coprophagous taxa can be further subdivided into those that prefer dung from non-ruminal vertebrates, ruminal vertebrates, or omnivorous vertebrates. (5) Species were therefore scored one point per dung type they are able to utilise (carrion, ruminal dung, omnivorous dung, and non-ruminant dung), and then, depending on how many sources a particular dung beetle species utilized, as indicated in the literature (Davis et al., 2020) and our own collections, a score between 1 and 4 was assigned to each species. A larger head area indicates that a beetle can manipulate relatively larger quantities of dung (Nervo et al., 2014; Tonelli et al., 2018). We therefore calculated (6) the head surface area of each dung beetle species, using Image J software (ImageJ2, National Institute of Health) after taking photos of dismembered beetles on a background with a standardised grid of 1 mm × 1 mm. Similarly, beetles with larger fore tibia will presumably manipulate dung more easily than those with smaller fore tibia, therefore (7) fore tibial area, as determined using Image J software, was also included in the analyses. Species with longer legs can generally move faster and roll larger dung balls than species with shorter legs (Inward et al., 2011). We therefore measured (8) hind tibia length (mm) using digital callipers. Long-distance dispersal ability of dung beetles is strongly affected by wing load (da Silva and Hernández, 2015a). Wing load (9) was calculated from the dry mass data and the area of the hind wings, which were determined using Image J software (mg/mm²). To calculate wing area, the wings were removed before drying of specimens, opened to their full extent, and photographed. Different dung beetle species may differ in terms of which specific months of the year they are active. Therefore, the final trait included was (10) phenology (seasonal activity). Phenology information was sourced from the literature (Davis et al., 2020) and according to our own collections and coded as outlined by Tonelli et al. (2017) for analyses.

Functional trait indices were calculated across different land uses, dung types, and landscape features using the *FD* package (Laliberté and Legendre, 2010) in R. Functional diversity indices assessed in the present study included FRic (functional richness) based on presence-absence data (Mason and De Bello, 2013), FEve (functional evenness) to evaluate how evenly ecological niches are occupied

(Villéger et al., 2008), and FDis (functional dispersion) to determine the degree of functional differences between species within communities. In addition, FDiv (functional divergence), a measure which is used to identify if resources are evenly exploited by the community (Mason and De Bello, 2013), and RaoQ (Rao's quadratic entropy (Q)), a calculation of the abundance-weighted variance of the dissimilarities between all species pairs (Botta-Dukát, 2005), were calculated.

All data sets generated for the different functional diversity indices were first tested for normality using a Shapiro–Wilk test in R (R Development Core Team, 2020). Where possible, data that were not normally distributed were transformed using the arcsine transformation, as was the case for FEve and FDiv data. If not, data were analysed untransformed, as was the case for FRic, FDis, and RaoQ. To compare differences in functional diversity indices regarding land use, landscape feature, and dung type, we used linear modelling procedures. Best-fit models were determined from full models (those containing all explanatory variables) with a stepwise forward selection method based on AICc values using the *AICcmodavg* package in R (Mazerolle, 2017). For data that could not be transformed (FRic, FDis, and RaoQ), generalized linear mixed-effect models with a gamma distribution and log link function, and site partitioned by the abovementioned precipitation gradient as a random effect, were calculated using the *lme4* package (Bates et al., 2014). For transformed data (FEve and FDiv), linear mixed models were used. Significance of variables retained after model forward selection procedures were tested using the *anova* function in R. Hereafter, the significant main effects were separated using Tukey post-hoc tests.

3. Results

A total of 2584 dung beetle individuals, comprising 15 species, were sampled (Table 1). Species estimates neared observed species richness (Chao2 = 17.00 ± 3.74; Jackknife2 = 17.99) (Table 2). In total, 601 dung beetle individuals were collected in the grazed areas, representing twelve species; 1084 dung beetle individuals were collected in recently non-grazed areas, representing thirteen species; and 899 dung beetle individuals were collected in non-grazed areas, consisting of thirteen species. Most species and individuals were collected in the flatlands and dry riverbeds. There were 1205 dung beetle individuals collected using pig dung, 1322 using cow dung, and 57 using sheep dung (Table 1).

3.1. Effects of land abandonment, dung type, and landscape feature on dung beetle diversity

Sheep dung always attracted significantly fewer dung beetle species,

Table 2

Observed and estimated overall dung beetle species richness and species richness associated with the different land use types, dung types and landscape features. Values in parentheses are SE.

	Observed	Chao2	Jackknife2
Overall	15	17.00 (3.74)	17.99
Land use			
Non-grazed	13	15.00 (3.74)	15.94
Recently non-grazed	13	13.25 (0.73)	15.96
Grazed	12	12.00 (2.86)	15.92
Dung type (bait)			
Pig	15	15 (5.83)	20.88
Cow	12	13 (3.08)	18.88
Sheep	5	9.5 (7.19)	9.92
Landscape feature			
Flatlands	14	15 (5.77)	20.87
Dry riverbeds	12	12 (2.64)	17.88
Rocky hills	11	15.5 (7.19)	15.92

individuals, and biomass than cow dung and pig dung (which were statistically similar; Table 3). Dung beetle abundance and richness were significantly higher at recently non-grazed and non-grazed sites than at currently grazed sites (Table 3). In contrast, dung beetle biomass was significantly higher at grazed and recently non-grazed sites than at non-grazed sites (Table 3). Flatlands always had the highest dung beetle species richness, abundance, and biomass (Table 3). The abundance and biomass of dung beetles was also significantly higher in dry riverbeds than at rocky hill sites. However, this pattern was not reflected in the richness of beetles as those from dry riverbeds and from rocky hills did not differ significantly (Table 3).

Community assemblage composition of the dung beetles did not differ between those collected using pig dung and those collected using cow dung (Table 4). Sheep dung collected a unique dung beetle assemblage. Dung beetle assemblage composition differed between all three land-use types (Table 4), but that from non-grazed and recently non-grazed areas were more similar than either were to those on grazed sites (Fig. 2). Flatlands had a unique assemblage when compared to rocky hills but was similar to dry riverbeds (Table 4). Dry riverbeds and rocky hills had similar assemblages (Table 4).

3.2. Effects of land abandonment, dung type, and landscape feature on dung beetle functional diversity

Values for traits used in functional diversity calculations are presented in Supplementary Table S1. For FRic, the variables landscape feature, land use, and dung type were retained after forward selection

Table 1

Species table comprising the total abundance of individuals collected in different land use types (G—grazed, RNG—recently non-grazed, NG—non-grazed), using different dung baits (P—pig, C—cow, and S—sheep) and from different landscape features (F—flatlands, R—rocky hills, and D—dry riverbeds).

Species	Total	Land use			Dung type			Biotope		
		G	RNG	NG	P	C	S	F	R	D
Aphodinae sp. 1	2	0	1	1	1	1	0	1	0	1
Aphodinae sp. 2	406	96	179	131	125	279	3	242	50	114
Aphodinae sp. 3	84	38	26	20	58	23	0	16	33	35
Aphodinae sp. 4	160	27	23	110	55	105	43	54	59	47
Cheironitis scabrosus	54	7	15	32	19	35	0	23	10	21
Epirinus aeneus	1177	122	522	533	550	584	0	479	202	496
Euoniticellus intermedius	1	1	0	0	1	0	0	1	0	0
Escarabaeus satyrus	48	9	16	23	33	15	2	33	3	12
Euonthophagus vicarius	5	1	4	0	5	0	0	5	0	0
Onitis confusus	1	0	0	1	1	0	0	0	1	0
Onthophagus fugitivus	191	16	155	20	115	76	0	51	65	75
Scarabaeus basuto	4	0	2	2	3	1	4	3	0	1
Scarabaeolus damarensis	16	4	6	6	6	10	0	14	1	1
Scarabaeolus karrooensis	102	32	61	9	38	60	0	87	1	14
Scarabaeus viator	333	248	74	11	195	133	5	266	4	63
Total	2584	601	1084	899	1205	1322	57	1275	429	880

Table 3

The effect of land use (NG—non-grazed, RNG—recently non-grazed, G—grazed), dung type (P—pig, C—cow, and S—sheep) and landscape feature (F—flatlands, R—rocky hills, and D—dry riverbeds) on the abundance, species richness and biomass of dung beetles. For post hoc analyses variables were arranged from highest to lowest median values and with > meaning the preceding factor(s) is significantly larger than those that follow, = factors are equal, and $\geq$ indicates that the last factor is significantly larger than the first, but equal to the second (** $p < 0.01$, * $p < 0.05$).

	Abundance		Species Richness		Biomass	
	χ^2	Pairwise	χ^2	Pairwise	χ^2	Pairwise
Dung type	1786.6 ***	C=P > S	248.57 ***	C=P > S	64.928 ***	C=P > S
Land use	148.46 ***	NG = RNG > G	15.022 ***	NG = RNG > G	6.2832 *	G = RNG > NG
Landscape feature	434.71 ***	F > D > R	36.746 ***	F > R = D	60.516 ***	F > D > R

Table 4

Comparisons between community assemblage composition of dung beetles at sites that differ in land use, landscape feature and dung type (bait) and including the interactions between them (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Dung type (P—pig, C—cow, and S—sheep) and landscape feature (F—flatlands, R—rocky hills, and D—dry riverbeds).

Diversity Index	df	Pseudo-F	Pairwise
Land use	2	25.3910***	all different
Dung type	2	35.6920***	P=C; P≠S; C≠S
Landscape feature	2	2,9248*	R≠F; F=D; D=R
Dung type x Land use	4	3,1381*	
Dung type x Landscape feature	4	1,1780	
Land use x Landscape feature	4	0,8853	

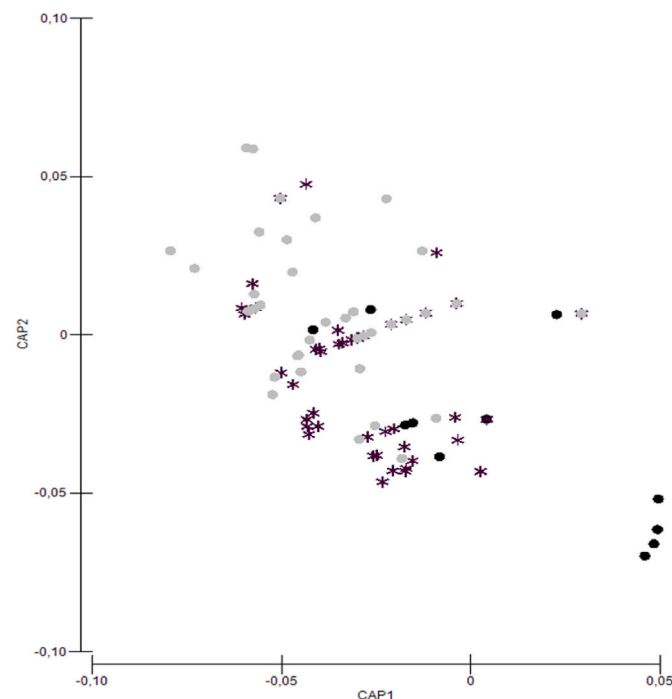


Fig. 2. Canonical analysis of principal coordinates (CAP) of the dung beetle assemblage for three different land use types (asterisks = non-grazed, grey dots = recently non-grazed, and black dots = grazed).

procedures (Table 5). For FEve, no variables were retained and for FDiv and FDis only landscape feature was retained. For RaoQ, landscape feature and dung type were retained in the model after forward selection procedures (Table 5). FEve, FDiv, FDis, and RaoQ were not affected by dung type. Differences in land use and dung had no effect on the functional evenness (FEve) or functional divergence (FDiv) of dung beetles. FDiv was affected by landscape feature, but the post-hoc testes could not discern which level of the factor was responsible for this (Table 5). Except for FRic, different land uses did not affect dung beetle functional

Table 5

Significant variables (landscape feature, land use, and dung type) that explain functional diversity (FRic—functional richness, FEve—functional evenness, FDiv—functional divergence, FDis—functional dispersion, RaoQ—Rao's quadratic entropy (Q)). For post hoc analyses variables were arranged from highest to lowest median values and with > meaning the preceding factor(s) is significantly larger than those that follow, = factors are equal, and $\geq$ indicates that the last factor is significantly larger than the first, but equal to the second. N/A – no variables retained after model selection procedures. Land use (NG—non-grazed, RNG—recently non-grazed, G—grazed), dung type (P—pig, C—cow, and S—sheep) and landscape feature (F—flatlands, R—rocky hills, and D—dry riverbeds).

Functional diversity index	Variables retained after forward selection	F/X ²	P	Post hoc
FRic	Dung type	125.661	0.0166	P=C > S
	Landscape feature	108.709	<0.001	F=D>R
	Land use	93.205	<0.001	RNG = NG $\geq$ G
FEve	N/A	N/A	N/A	N/A
FDiv	Landscape feature	3.1548	0.047	F = D = R
FDis	Landscape feature	6.7043	0.002	F = D > R
RaoQ	Dung type	2.5989	0.080	N/A
	Landscape feature	8.3529	<0.001	F=D>R

diversity. Non-grazed sites had the highest FRic and grazed sites the lowest, with non-grazed sites at intermediate levels (Table 5). Landscape features differed significantly in terms of dung beetle FRic, FDis, and RaoQ (Table 5). Flatlands and dry riverbeds had dung beetle communities with similar functional values, which were significantly higher than for those on rocky hills. Dung beetle functional diversity was similar for all dung types in terms of all diversity indexes except for FRic. Dung beetles attracted to pig dung and cow dung were similar in terms of FRic, and these were significantly higher than for the dung beetles attracted to the sheep dung (Table 5).

4. Discussion

We examined the influence of rangeland abandonment, dung type, and landscape features on dung beetle biodiversity and functional diversity in the semi-desert Nama-Karoo biome of South Africa. Like with other studies, it was shown that maintaining a mosaic of different landscape features is important for maintaining dung beetle biodiversity (Pryke et al., 2013, 2016) and functional diversity (Verdú et al., 2007). However, contrary to expectations, dung beetle richness, abundance, and functional richness were higher in areas where livestock activities were no longer practiced. This was seemingly driven by low numbers of dung beetles associated with the pelleted dung of sheep, the dominant dung resource on active farms. Instead, there were greater attraction to dung resources from omnivores and herbivores that provide dung with wetter consistency. Large native herbivores are still largely excluded from non-grazed areas, but the availability of dung resources from carnivores (e.g., jackal and caracal), omnivores (e.g., baboons), and other smaller mammals will be higher due to decreased pressures on their populations (Woodgate et al., 2018). Therefore, patterns of increased dung beetle biodiversity and functional richness on

abandoned farmland in this region seems to be driven by increased high-quality resource availability due to decreased pressures on native vertebrates (Nependa et al., 2021).

A study on dung beetle diversity in a Brazilian semi-arid ecosystem indicated that dung beetle communities in undisturbed areas were species poor in comparison to those that were actively farmed (Liberal et al., 2011). In semi-arid central Spain, Lobo et al. (2006) and Tonelli et al. (2018) found that the abandonment of traditional pastoral systems caused decreased dung beetle diversity. Similarly, Verdú et al. (2007) found that grazing by livestock encouraged dung beetle diversity in Mexico. The difference between these studies and the present one seems to be that dung beetles were able to readily exploit the exotic dung resources produced by livestock in the former. Differences in dung quality and not quantity may therefore be the most important key to disentangle these contrasting patterns (Nependa et al., 2021). To improve conditions for enriched dung beetle diversity and function, reintroductions of native meso-herbivores, such as wildebeest and zebra, should be considered in the abandoned Nama-Karoo areas (Lovegrove and Siegfried, 1993). Increased dung beetle diversity and functional richness will also benefit active farming areas as it improves vegetation development, nutrient cycling, and parasite and pest control (Nichols et al., 2008). Therefore, increased abundance of omnivores, ruminants, and wild herbivores on farms should be encouraged.

The increase in functional richness followed an increase in species richness at abandoned sites as a higher number of species can fill more niches when there is not considerable niche overlap between taxa (Slade et al., 2007; Tonelli et al., 2018). This increase will improve ecological buffering against environmental fluctuations (Tilman et al., 2014) and provide invasion resistance to more generalist dung beetle species (Dukes, 2001; Tonelli et al., 2017). However, functional richness does not incorporate relative abundance, and rare species with extreme trait values can greatly inflate functional richness. This was evident in the present study where rare and functionally divergent taxa, such as *Euoniticellus intermedius*, *Euonthophagus vicarius*, and *Onitis confusus*, added considerably to increased functional richness. Diversity measures that accounted for abundance (FEve, FDiv, FDis, and RaoQ) showed no difference between the different land-use types. Most ecological niches across the different land-use types are therefore regularly filled, the functional diversity is similar between land-use types when these rare taxa are not present, and resources are evenly exploited by the community. Rarity of taxa that drive divergence between sites is likely caused by rarity in their favoured resources, linked to diminished numbers of native mammals, and thus reintroductions should lead to increases in their numbers (Laliberté and Legendre, 2010). For example, all three the above-mentioned species were attracted to omnivore dung, a resource that is limited on active farms due to hunting and the control of animals such as honey badger (*Mellivora capensis*), baboon (*Papio ursinus*), and jackal (*Lupulella mesomelas*) (Somers et al., 2018).

Despite the generally low richness and abundance of dung beetles at sites that are still grazed, dung beetle biomass was still high at grazed sites, indicating that relatively larger dung beetle species can become locally rare after land abandonment (Tonelli et al., 2018). Larger species, such as *Scarabaeus viator* were common in the grazed areas and need more dung resource for feeding, nesting, and reproduction than smaller taxa. Their reduced numbers in abandoned rangelands may therefore indicate a scarcity of resources for reproduction. Once sheep are removed, smaller dung beetle species increased in numbers, as they require less dung for reproduction, which makes them more opportunistic and less vulnerable to changes in the quantity of dung resources (Nichols et al., 2008; Tonelli et al., 2018). Dung quality from livestock may be further reduced by using veterinary medical products which affects dung beetle communities (Scholtz et al., 2009). Although not tested, larger species may be more sensitive to veterinary medical products than the smaller species encountered in this study. This reduction in the average dung beetle size after land abandonment will affect dung beetle functional roles (Nependa et al., 2021). Therefore, to

maintain larger dung beetles and their associated ecological functions, numbers of native meso-herbivores should be increased after land abandonment. Not only will these increase the availability of key dung resources for dung beetles, but they will also increase the resources for carnivorous and omnivorous mammals, creating a positive feedback loop between vertebrate functions and dung beetle functions in this sensitive ecosystem.

Dung beetles are influenced by several environmental characteristics, including elevation, vegetation structure, as well as soil type and moisture (Davis et al., 2020). The existence of unique communities associated with the different biotopes is therefore not unexpected. Flatlands and dry riverbeds generally had significantly higher dung beetle abundance, richness, biomass, and functional diversity than rocky hills. Therefore, more ecological niches (and more divergent niches) are exploited in the flatlands and dry riverbeds. The hills have rocky soils, which makes them less habitable to dung beetles, leading to decreased availability of niches in this biotope. Unfortunately, the flatland and dry riverbed habitats are also most severely impacted by farming activities as these areas are often overstocked, leading to overgrazing by sheep and reduced moisture retention (Li et al., 2016). As sheep dung is not a favoured resource for dung beetles in our study area, larger ecosystem impacts (e.g., decreased secondary seed dispersal) and decreased ecosystem services (reduced clearing of pest breeding sites) can result. Our results do however contrast to what has previously been found in a South African dryland ecosystem in that sheep dung performed favourably to other dung types, including cattle- and pig dung, in the Tswalu Kalahari Reserve (Davis et al., 2010). Dung beetle species composition and vertebrate species composition are however different between these two regions and the effect of sheep farming on dung beetle communities may therefore substantially differ between different regions.

To conclude, increased functional richness in abandoned rangelands was closely linked to increased dung beetle richness. The link between species richness and increased functional diversity is expected as different species fill different niches in ecological systems, leading to improved ecosystem functioning (Petchey and Gaston, 2002; Tonelli et al., 2018). However, whether the increases reported in the current study translates into increased ecosystem functions, such as dung removal and secondary seed dispersal, is not known. There is therefore a strong need for future studies to incorporate more functional measures related to changes in dung beetle diversity because of anthropogenic change on ecosystem processes. Nonetheless, rehabilitation of these abandoned rangelands by introduction of the appropriate native mammals, and subsequently managing the area as a protected reserve, would promote the presence and maintenance of vital dung beetles. The same principle would also be true for farmers in the matrix, such as mixed-animal farming and less predator control where possible. In addition, rainfall patterns are known to strongly influence dung beetle communities in the study area with peak numbers corresponding with peak rainfall during late summer (Davis et al., 2016). Our sampling (Spring and winter) was therefore not conducted during peak activity season. Whether the additional taxa and numbers that are active during the peak activity season will show different patterns of associations with dung, landscape features and land use remains to be determined.

CRedit authorship contribution statement

Adam Steed: Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Francois Roets:** Conceptualization, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Gabriella J. Kietzka:** Formal analysis, Visualization, Writing – review & editing. **Casparus J. Crous:** Conceptualization, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jaridenv.2022.104738>.

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